Kernix 0.11.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



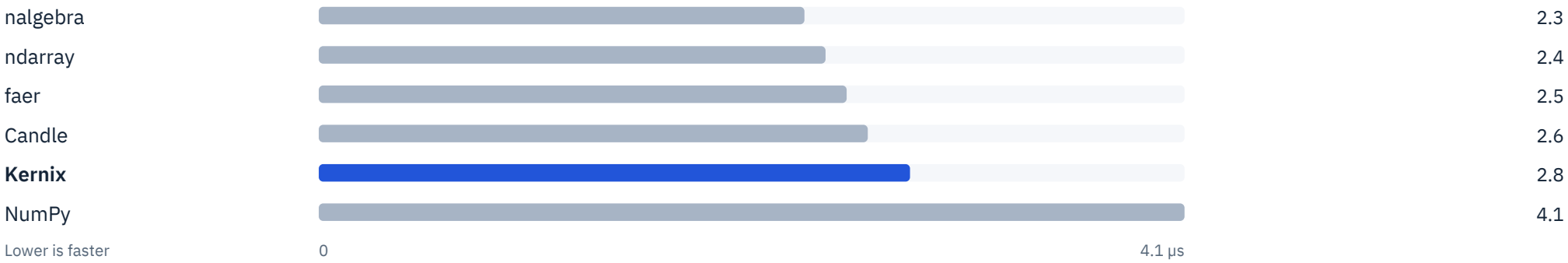
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



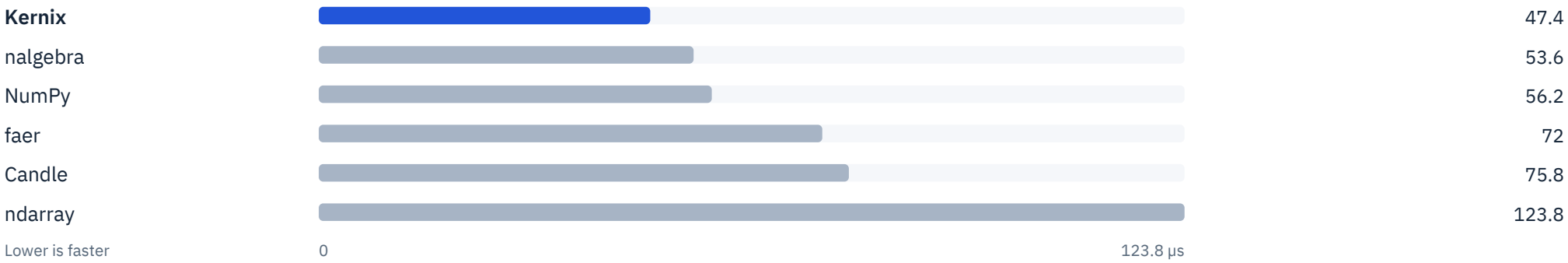
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded

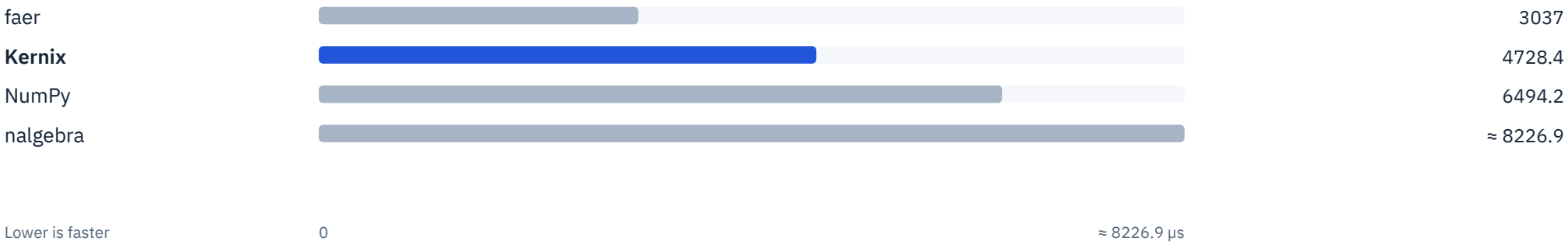


Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



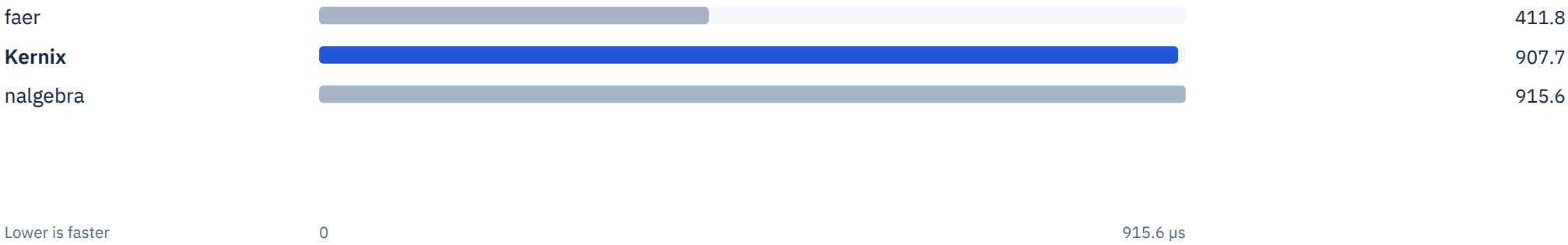
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



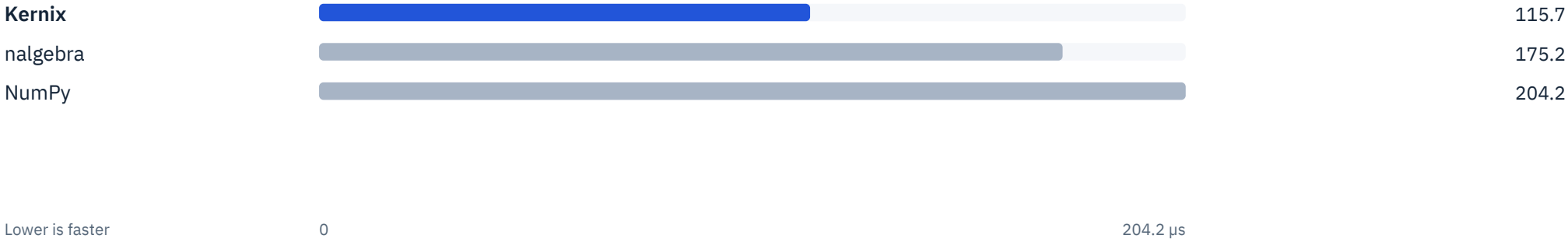
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

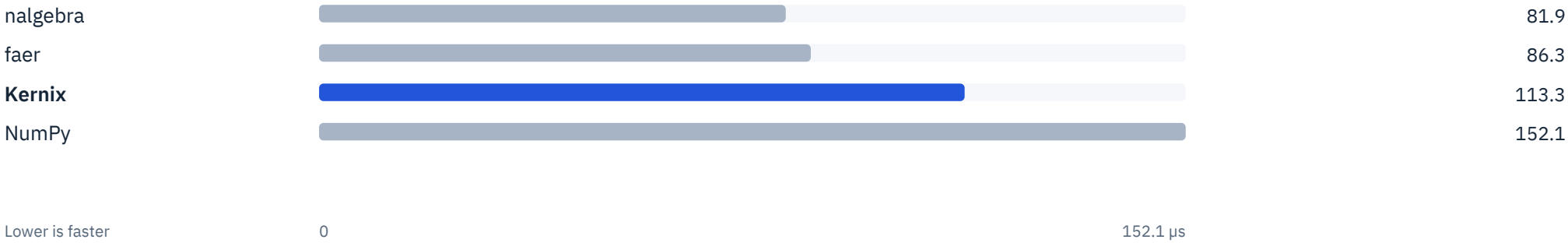
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output

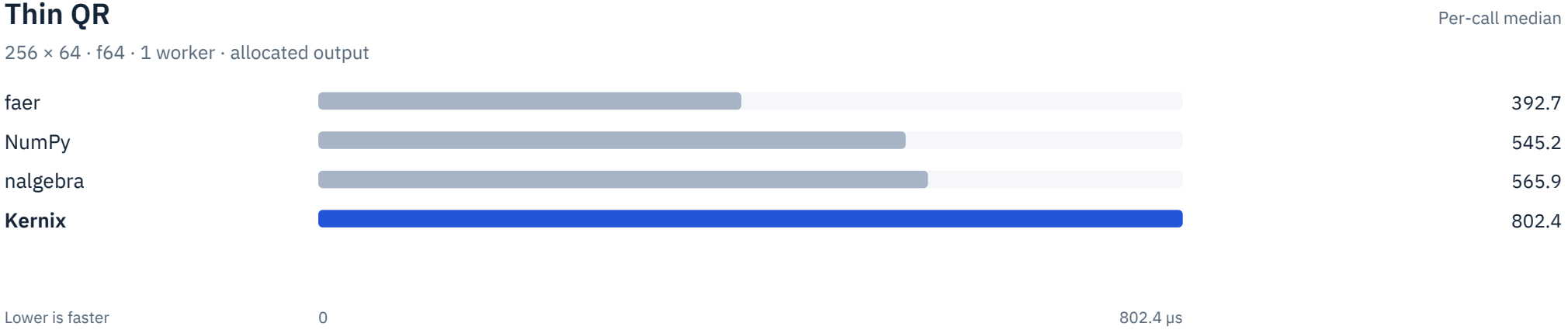


Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

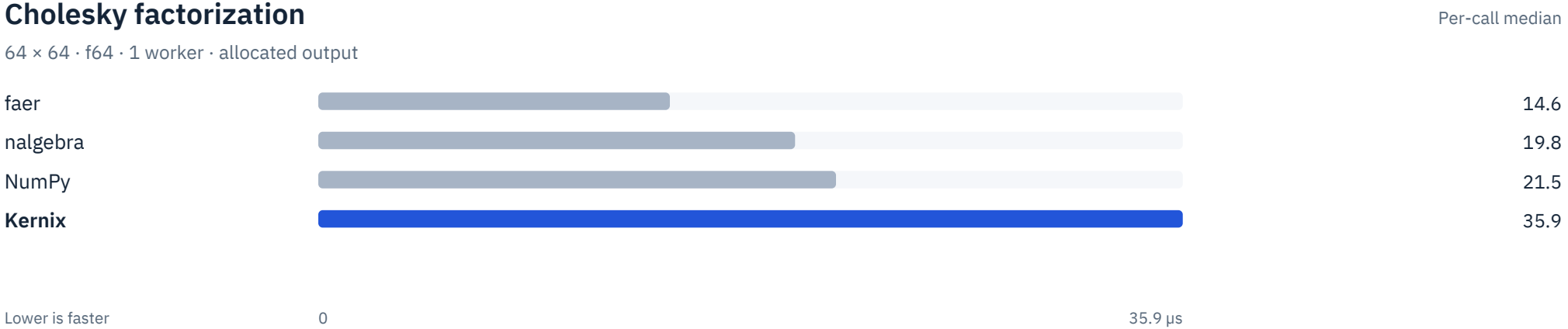
Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



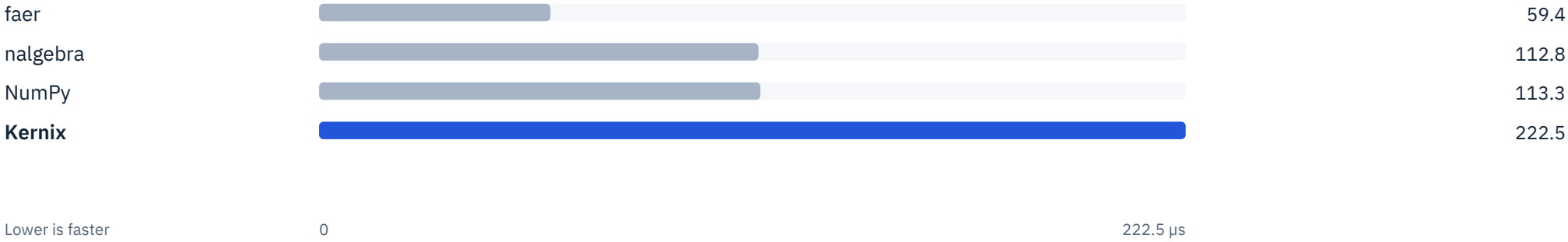
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

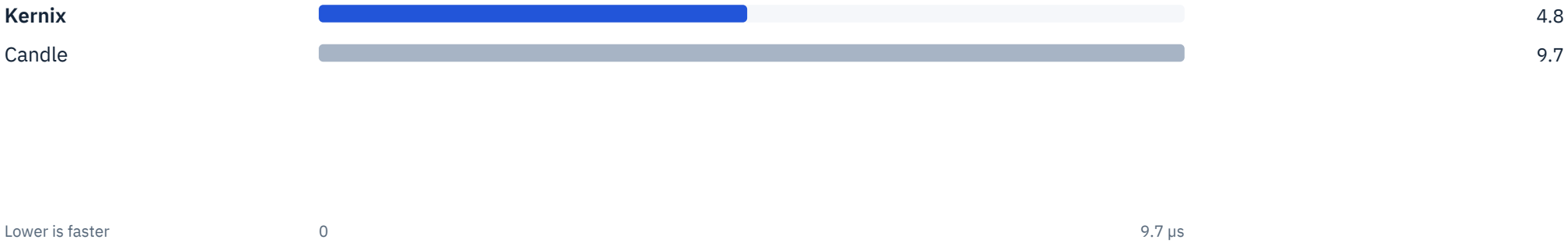
sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

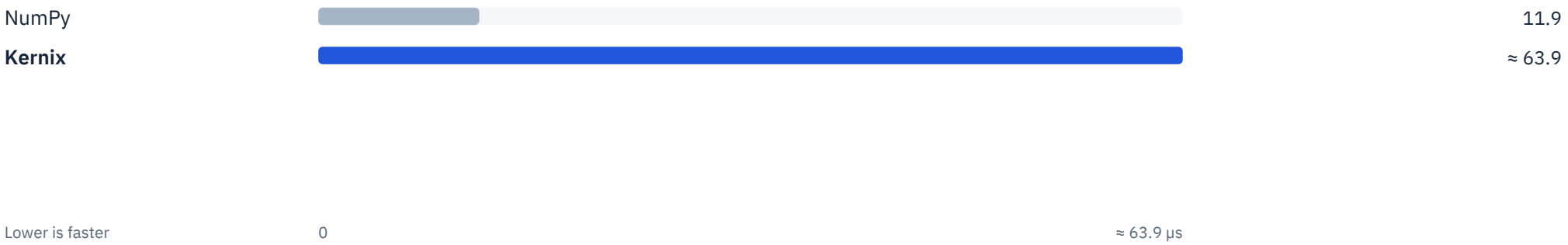


Kernix 0.11.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

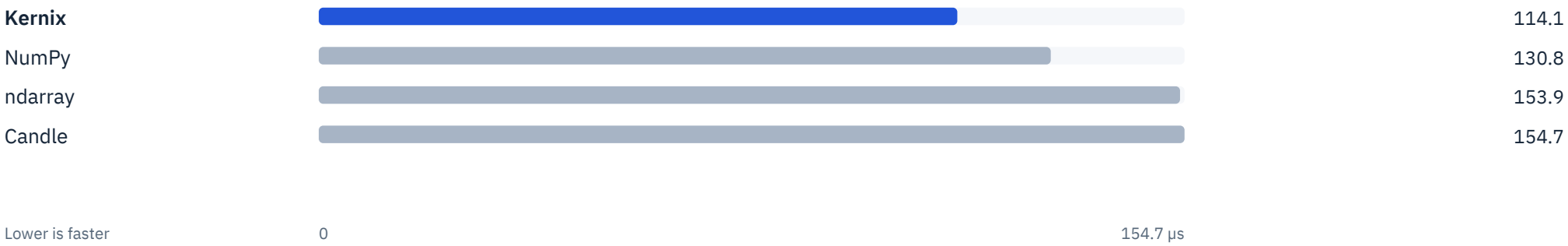


Kernix 0.11.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



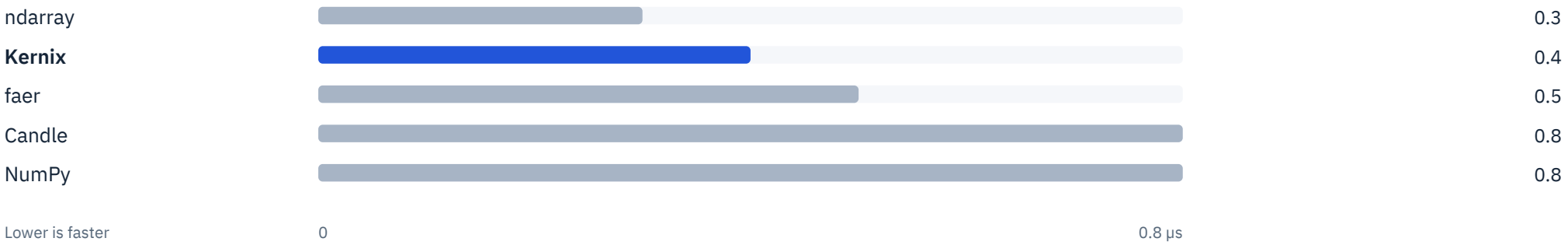
Kernix 0.11.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9947257383966245	0.9851694915254238 – 1.0042328042328041

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.997872340425532	0.6146788990825688 – 1

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0014998636487593	0.8458338148618976 – 1.0089151008092168

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0002831257078142	0.9943804439449283 – 1.0014124293785311

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9985915492957746	0.9950980392156863 – 1.4880581516095535

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	0.9974059662775616 – 1.0039011703511054

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0076957899502037	0.972101133391456 – 1.0108790252393385

Cholesky factorization

256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.695573074590661	0.6864988558352402 – 0.7127808988764045

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.5457797784753895	0.5446214676125131 – 0.5541659264523006

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	0.9640371229698376 – 1.0133010882708586

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9986489529385274	0.9724872420678944 – 1.0055121727147451

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.008695652173913	0.9948186528497409 – 1.0373001776198933

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	142400	142.4
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	101500	101.5
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	56400	56.4
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	75200	75.2
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2600	2.6
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4100	4.1
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2300	2.3
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	60900	60.9
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14100	14.1
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13600	13.6
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	58600	58.6
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15900	15.9
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14600	14.6
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	101300	101.3
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	141300	141.3
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	116000	116
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	120300	120.3

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	128400	128.4
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	141000	141
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	900100	900.1
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	769700	769.7
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	918300	918.3
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1176400	1176.4
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	860000	860
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1069100	1069.1
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	29700	29.7
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7400	7.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7100	7.1
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	30200	30.2
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7100	7.1
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10400	10.4
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	47400	47.4
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	123800	123.8
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	75800	75.8
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56200	56.2
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53600	53.6
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	72000	72

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	430500	430.5
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	389500	389.5
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	394800	394.8
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	431500	431.5
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	415800	415.8
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	396300	396.3
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4000	4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5000	5
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	5200	5.2
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	39500	39.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	84100	84.1
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	441000	441
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	57500	57.5
Cholesky factorization / nalgebra 512 × 512 · f64	8226900.000000001	8226.900000000001
Cholesky factorization / NumPy 512 × 512 · f64	6494200	6494.2
Cholesky factorization / faer 512 × 512 · f64	3037000	3037
Cholesky factorization / Kernix 512 × 512 · f64	4728400	4728.4
Column-pivoted QR / Kernix 96 × 32 · f64	125100	125.1
Column-pivoted QR / faer 96 × 32 · f64	78000	78
Column-pivoted QR / nalgebra 96 × 32 · f64	107700	107.7

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	411800	411.8
Column-pivoted QR / nalgebra 256 × 64 · f64	915600	915.6
Column-pivoted QR / Kernix 256 × 64 · f64	907700	907.7
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	105100	105.1
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	113300	113.3
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	101800	101.8
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	175200	175.2
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	204200	204.2
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	115700	115.7
Thin QR / Kernix 96 × 32 · f64	113300	113.3

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	86300	86.3
Thin QR / NumPy 96 × 32 · f64	152100	152.1
Thin QR / nalgebra 96 × 32 · f64	81900	81.9
Thin QR / faer 256 × 64 · f64	392700	392.7
Thin QR / Kernix 256 × 64 · f64	802400	802.4
Thin QR / NumPy 256 × 64 · f64	545200	545.2
Thin QR / nalgebra 256 × 64 · f64	565900	565.9
Cholesky factorization / NumPy 64 × 64 · f64	21500	21.5
Cholesky factorization / nalgebra 64 × 64 · f64	19800	19.8
Cholesky factorization / faer 64 × 64 · f64	14600	14.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	35900	35.9
Cholesky factorization / Kernix 128 × 128 · f64	222500	222.5
Cholesky factorization / nalgebra 128 × 128 · f64	112800	112.8
Cholesky factorization / faer 128 × 128 · f64	59400	59.4
Cholesky factorization / NumPy 128 × 128 · f64	113300	113.3
Cholesky factorization / faer 256 × 256 · f64	315300	315.3
Cholesky factorization / Kernix 256 × 256 · f64	737700	737.7
Cholesky factorization / nalgebra 256 × 256 · f64	842000	842
Cholesky factorization / NumPy 256 × 256 · f64	985100	985.1
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	59400	59.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	13700	13.7
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4800	4.8
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	9700	9.7
Model inference / Candle batched normalized/sliced MLP · f64	8200	8.2
Model inference / Kernix batched normalized/sliced MLP · f64	3500	3.5
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4800	4.8
Model recording + forward / Candle batched normalized/sliced MLP · f64	10100	10.1
Model backward / Kernix prepared graph → fresh gradients · f64	5700	5.7
Model backward / Candle prepared graph → fresh gradients · f64	31500	31.5
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63899.99999999999	63.89999999999999
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	11900	11.9
Row softmax / Candle 256 × 64 · f64	154700	154.7
Row softmax / ndarray 256 × 64 · f64	153900	153.9
Row softmax / Kernix 256 × 64 · f64	114100	114.1
Row softmax / NumPy 256 × 64 · f64	130800	130.8
Create zeros / Kernix 64 × 64 · f32	400	0.4
Create zeros / faer 64 × 64 · f32	500	0.5
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create zeros / Candle 64 × 64 · f32	800	0.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	800	0.8
Create ones / Candle 64 × 64 · f32	700	0.7
Create ones / NumPy 64 × 64 · f32	2100	2.1
Create ones / Kernix 64 × 64 · f32	600	0.6
Create ones / ndarray 64 × 64 · f32	400	0.4
Create ones / faer 64 × 64 · f32	500	0.5